

OPEN HARDWARE SPARKS INNOVATION IN ROBOTICS

There are many issues bothering robotics in India—lack of advanced components, insufficient know-how to patch these together into finished products, high cost of parts and platforms, and so on. A touch of Open Source could be the magical boost that the Indian robotics community needs



DARwin-OP Open Source robot (Image courtesy: <http://robotarime.com>)

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If you speak to robotics enthusiasts in India, they are sure to complain about dearth of low-cost components, lack of affordable but high-quality finished products, insufficient knowledge to identify the right components for an application and put them together efficiently, scarce support for complex products and so on.

We set out to explore whether Open Source hardware (OSH) could be of any help, and were in for a pleasant surprise as several members of the community felt that it could not only sort out the existing problems but also improve the robotics scene in India in

many other ways.

Problems aplenty

The robotics community faces different problems at different levels. Students, for example, seem to be affected by the unavailability of components.

Praveen Pitchai, a member of RoboMSR, a group of robotics enthusiasts who are being mentored by Dr K.G. Srinivasa at M.S. Ramaiah Institute of Technology, Bengaluru, says, "There is a dearth of reliable and low-cost components manufactured in India. It is difficult to compete with the frontrunners in the field of robotics due to this very reason. Once a new component like a sensor or a

microcontroller is found and acquired from vendors outside India, it becomes difficult to configure it to the required specifications within a project as there are set standards to be followed by the company that manufactures them. This increases not only the costs involved in a project but also its complexity."

Rohit de Sa, nucleus member, Centre for Robotics and Intelligent Systems, BITS-Pilani, adds, "Components in raw form (resistors, capacitors, chips) are available. However, good-quality finished products (for example, a microcontroller development board or a long-range radio frequency transceiver) are difficult to find."

On the other hand, Fahad Azad, managing partner of Robosoft Systems, feels that the lack of components by itself is not the main problem. "It has never been only the lack of components in India for most of the cases but also lack of skills to identify the components required in an application. Even when the components are identified, they are out of reach due to economic factors. Thus the *jugaad* mode kicks in," he says.

Open Source could be the panacea

All three agree that OSH or open hardware, in some form or the other, can sort out these problems.

OSH is basically the concept of Open Source (as popularised by Open Source software) applied to the design and development of hardware components. Open hardware is not free-of-cost. It merely refers to the hardware for which the underlying technology or building blocks—the design and software—are